

TUGAS ELEKTRONIKA

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①

Jawab:

$$R_s = R_2 + R_3 = 2\Omega + 2\Omega = 4\Omega$$

$$R_p = \frac{R_2 \times R_3}{R_2 + R_3} = \frac{2 \times 2}{2 + 2} = 1\Omega$$

$$R_{tot} = R_1 + R_p = 1 + 1 = 2\Omega$$

②

Jawab:

$$R_s = R_2 + R_3 + R_4 = 3\Omega + 3\Omega + 3\Omega = 9\Omega$$

$$R_p = \frac{R_2 \times R_3 \times R_4}{R_2 + R_3 + R_4} = \frac{3 \times 3 \times 3}{3 + 3 + 3} = 1\Omega$$

$$R_{tot} = R_1 + R_p + R_5 = 1 + 1 + 1 = 3\Omega$$

③

Jawab:

$$R_s = R_2 + R_3 = 6\Omega + 9\Omega = 15\Omega$$

$$R_p = \frac{R_2 \times R_3}{R_2 + R_3} = \frac{6 \times 9}{6 + 9} = 3.6\Omega$$

$$R_{tot} = R_1 + R_p + R_4 = 3 + 3.6 + 5 = 11.6\Omega$$

④

Jawab:

$$R_p = \frac{R_2 \times R_3}{R_2 + R_3} = \frac{4 \times 6}{4 + 6} = 2.4\Omega$$

$$R_s = R_1 + R_p + R_4 = 2 + 2.4 + 5 = 9.4\Omega$$

⑤

Jawab:

$$R_p = \frac{R_2 \times R_3}{R_2 + R_3} = \frac{4 \times 12}{4 + 12} = 3\Omega$$

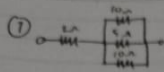
$$R_s = R_1 + R_p + R_4 = 10 + 3 + 7 = 20\Omega$$

⑥

Jawab:

$$R_p = \frac{R_2 \times R_3}{R_2 + R_3} = \frac{5 \times 15}{5 + 15} = 3.75\Omega$$

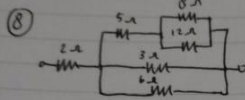
$$R_{tot} = R_1 + R_p + R_4 = 10 + 3.75 + 5 = 18.75\Omega$$



Jawab:

$$\bullet R_p = \frac{1}{\frac{1}{5} + \frac{1}{10} + \frac{1}{10}} = \frac{2}{\cancel{5}} = 2,5 \Omega$$

$$\bullet R_{tot} = 2 + 2,5 = 4,5 \Omega$$



Jawab:

$$\bullet R_{p_1} = \frac{8 \times 12}{8 + 12} = \frac{96}{20} = \frac{24}{5} = 4,8 \Omega$$

$$\bullet R_s = 5 + 4,8 = 9,8 \Omega$$

$$\bullet \frac{1}{R_{p_2}} = \frac{1}{9,8} + \frac{1}{3} + \frac{1}{6} = \frac{59}{98} \Rightarrow R_{p_2} = \frac{98}{59} = 1,661 \Omega$$

$$\bullet R_{tot} = 2 + 1,661 = 3,661 \Omega //$$